STABILIZATION OF THE CHAIN A[F^N] EXHIBITS THE PRO-OBJECT A/(F^N) AND THE PRO-OBJECT OF THE MULTIPLICATION MAPS AS ISOMORPHIC IN THE DERIVED CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The author writes 'We omit the calculation of compositions needed to show the statement on pro objects.'; that sentence is kept verbatim and is the one step left to the reader.

Allowed standard prerequisites (editorial).

- The derived category D(A) and the complex (A –f–> A) placed in degrees 0 and 1. the notation (A –f^n–> A) in the statement at source lines 26786-26789. Fixes the target of the two required arrows and the sense in which the pro-objects are compared.
- Stabilization of the chain of annihilator ideals A[f^n]. the hypothesis at statement lines 26783-26785. Provides c such that A/A[f^c] can be used in the commuting square, which is what makes the second arrow exist.

Lemma 0.2. *Let A be a ring. Let $f \in A$. If there exists an integer $c \geq 1$ such that $A[f^c] = A[f^{c+1}] = A[f^{c+2}] = \ldots$ (for example if A is Noetherian), then for all $n \geq 1$ there exist maps*

$$(A \xrightarrow{f^n} A) \longrightarrow A/(f^n), \quad \text{and} \quad A/(f^{n+c}) \longrightarrow (A \xrightarrow{f^n} A)$$

in $D(A)$ inducing an isomorphism of the pro-objects $\{A/(f^n)\}$ and $\{(f^n : A \to A)\}$ in $D(A)$.

Proof. The first displayed arrow is obvious. We can define the second arrow of the lemma by the diagram

$$\begin{array}{ccc} A/A[f^c] & \xrightarrow[f^{n+c}]{} & A \\ {\scriptstyle f^c}\downarrow & & \downarrow{\scriptstyle 1} \\ A & \xrightarrow{f^n} & A \end{array}$$

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Since the top horizontal arrow is injective the complex in the top row is quasi-isomorphic to $A/f^{n+c}A$. We omit the calculation of compositions needed to show the statement on pro objects. $\square$